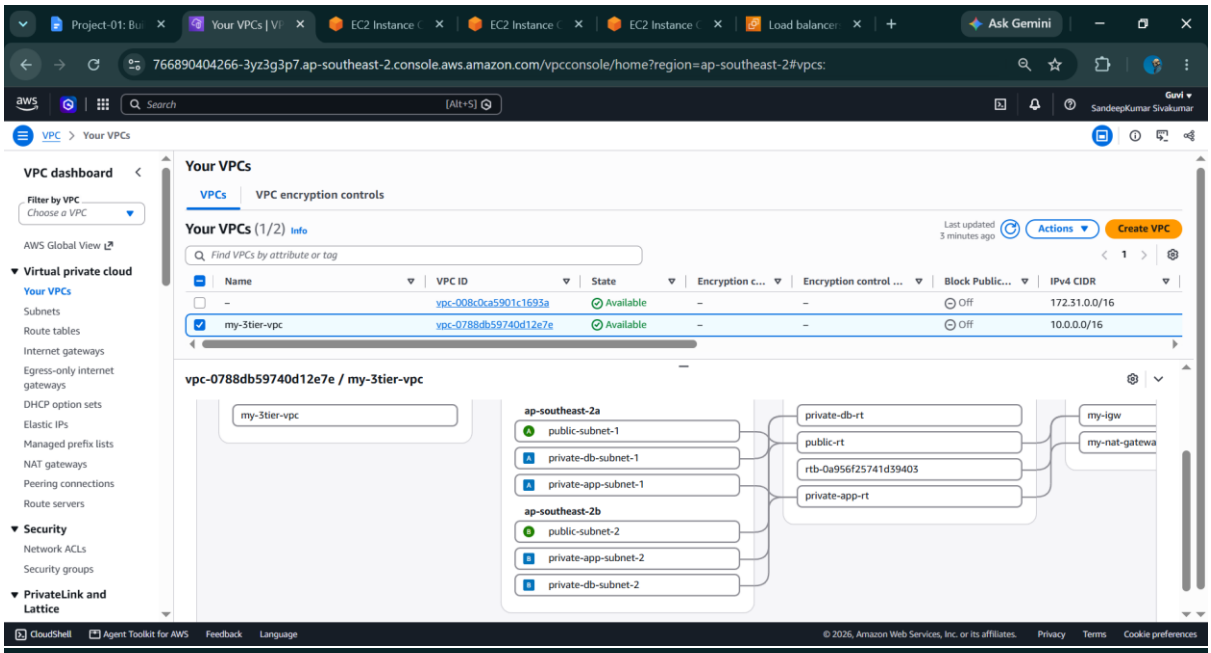
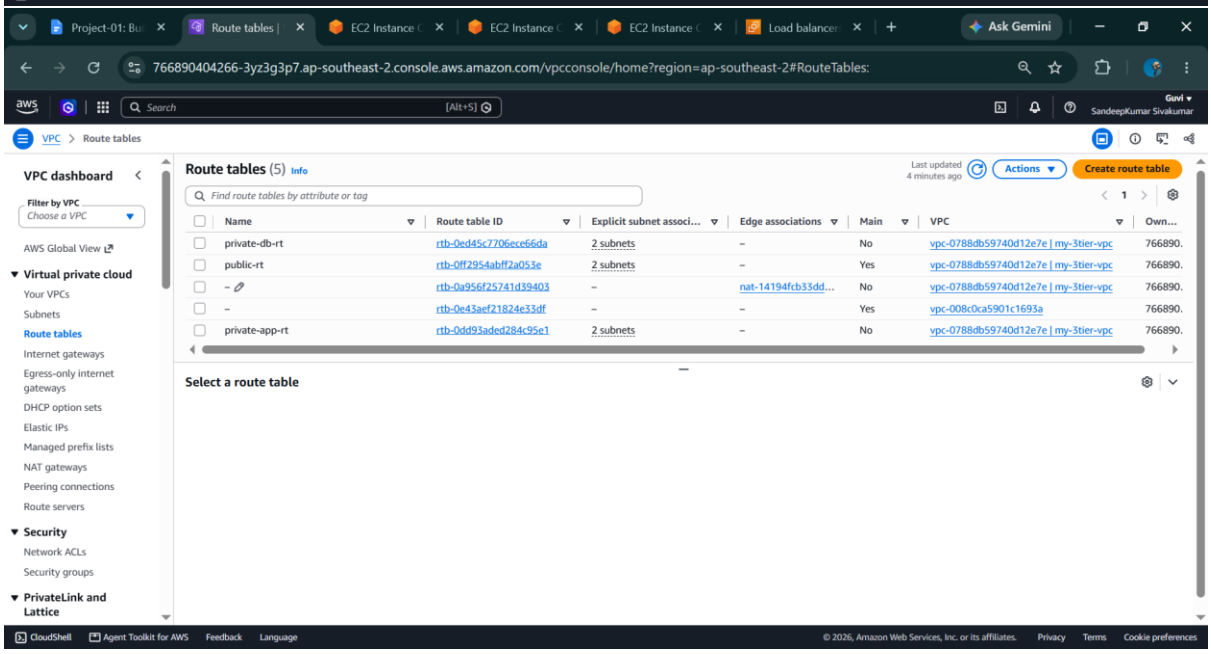
Project: Three-Tier-Architecture Web App

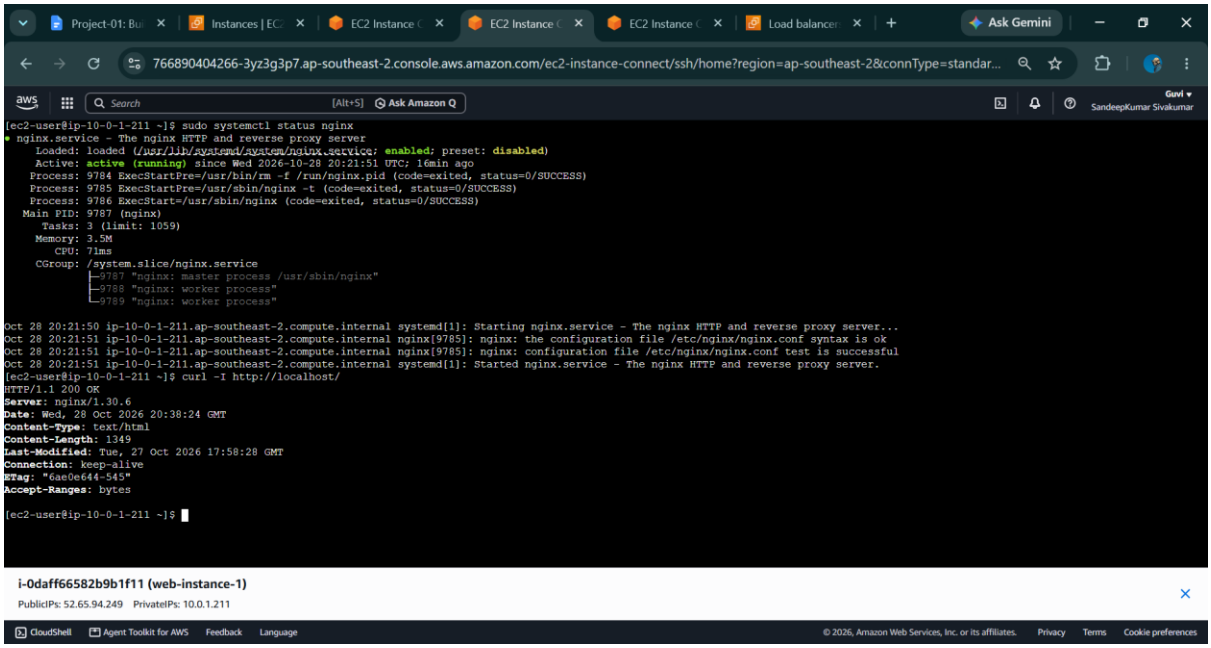
Application link: http://public-web-alb-1365904800.ap-southeast-2.elb.amazonaws.com/

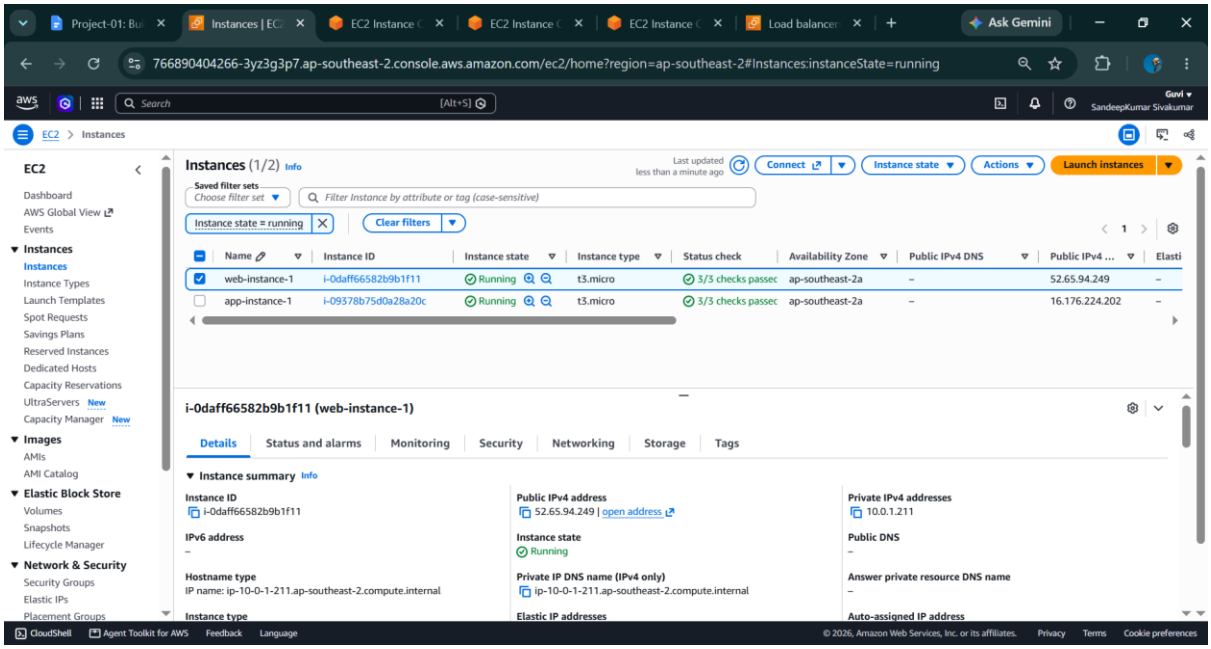
Task 1: Design and Configure the AWS Networking Infrastructure



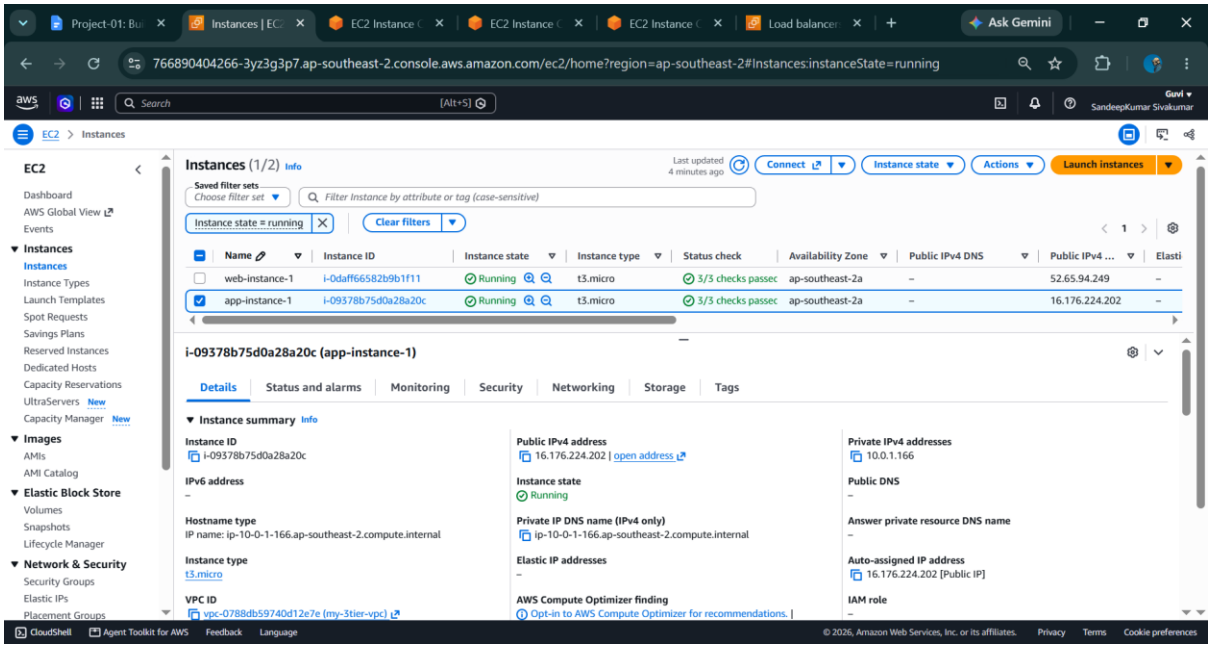


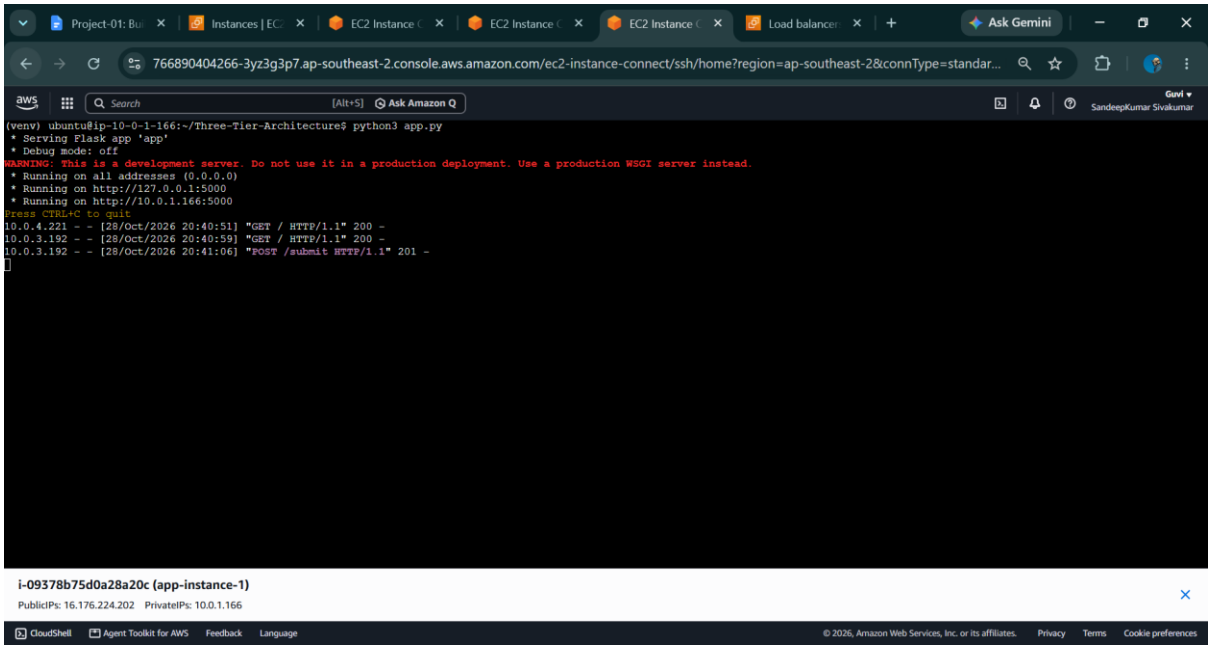
Task 2: Deploy the Web Tier (Presentation Layer)



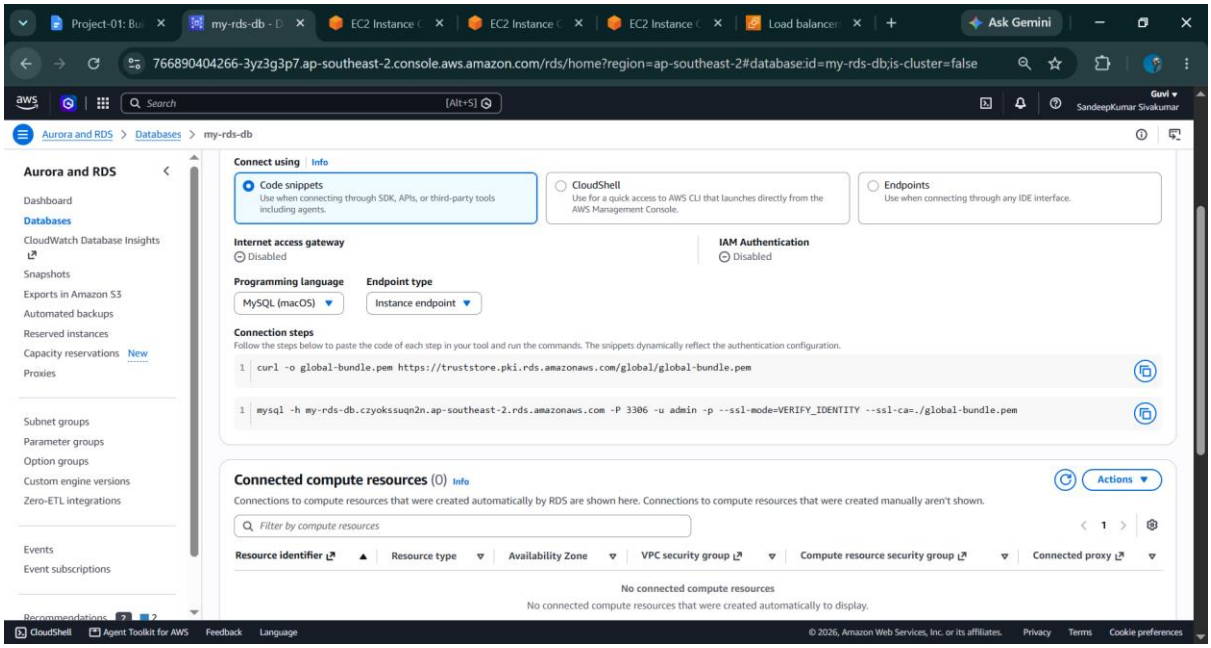


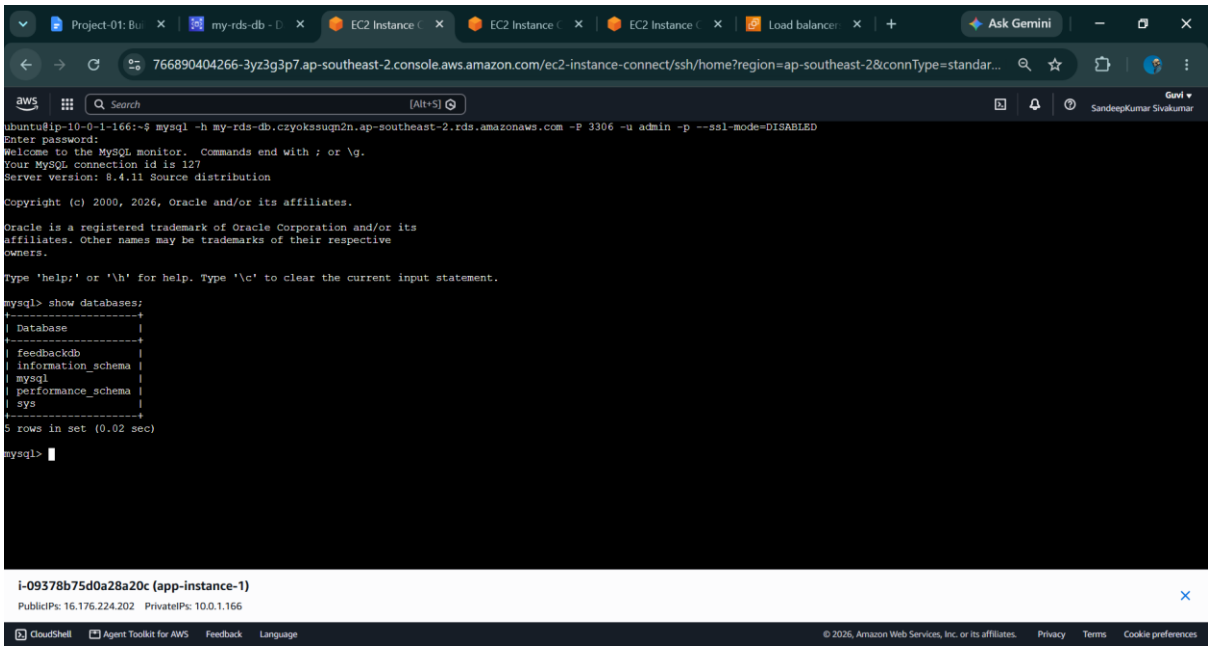
Task 3: Deploy the Application Tier (Backend API)



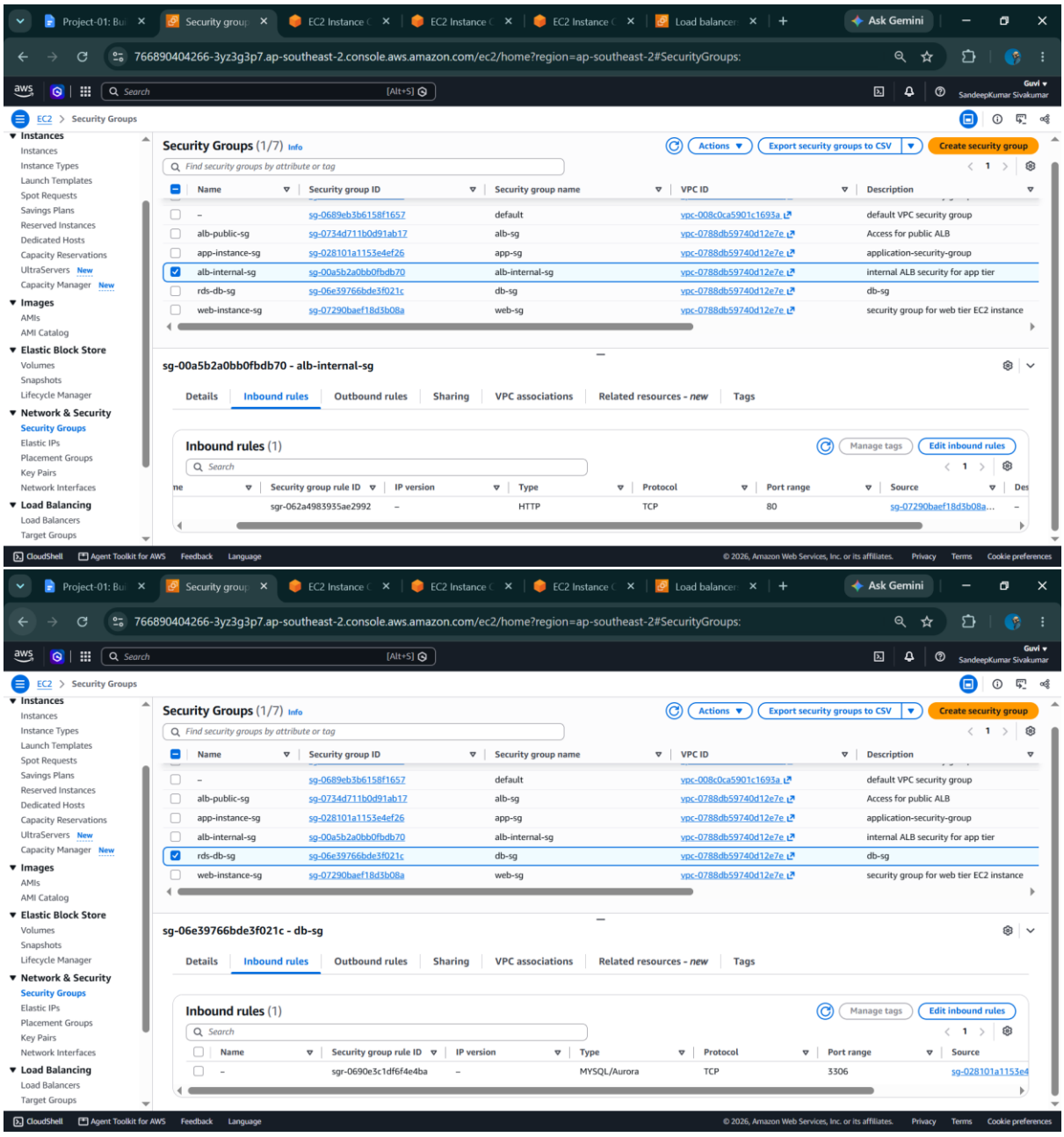


Task 4: Deploy and Configure the Database Tier

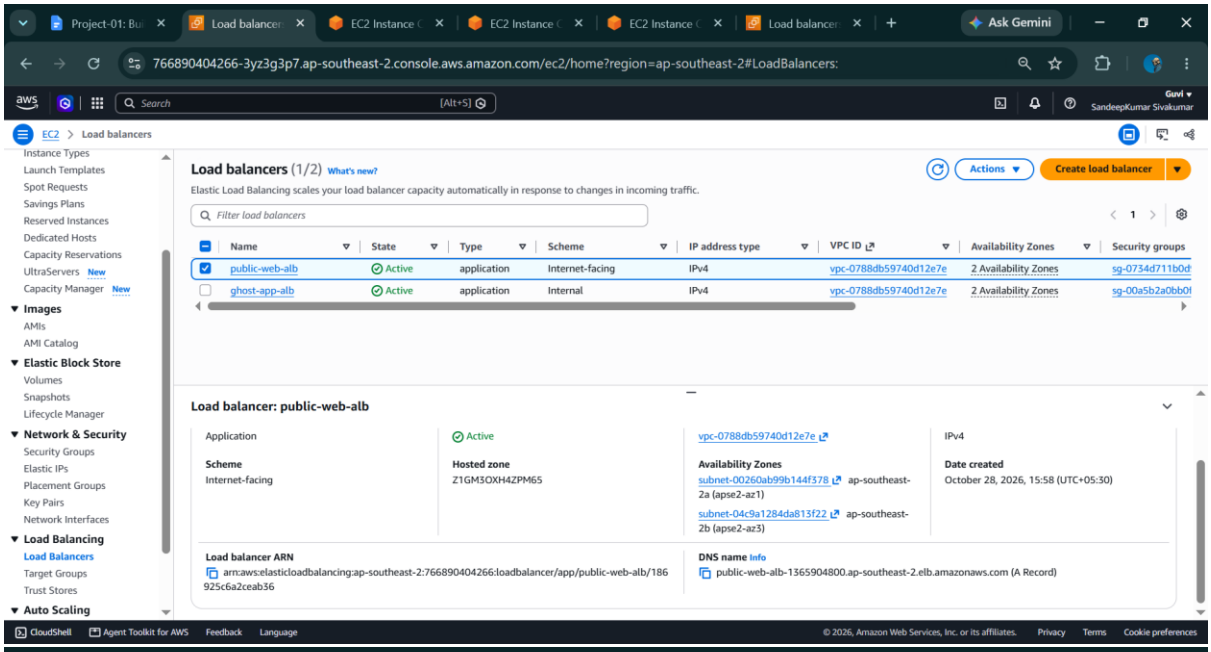


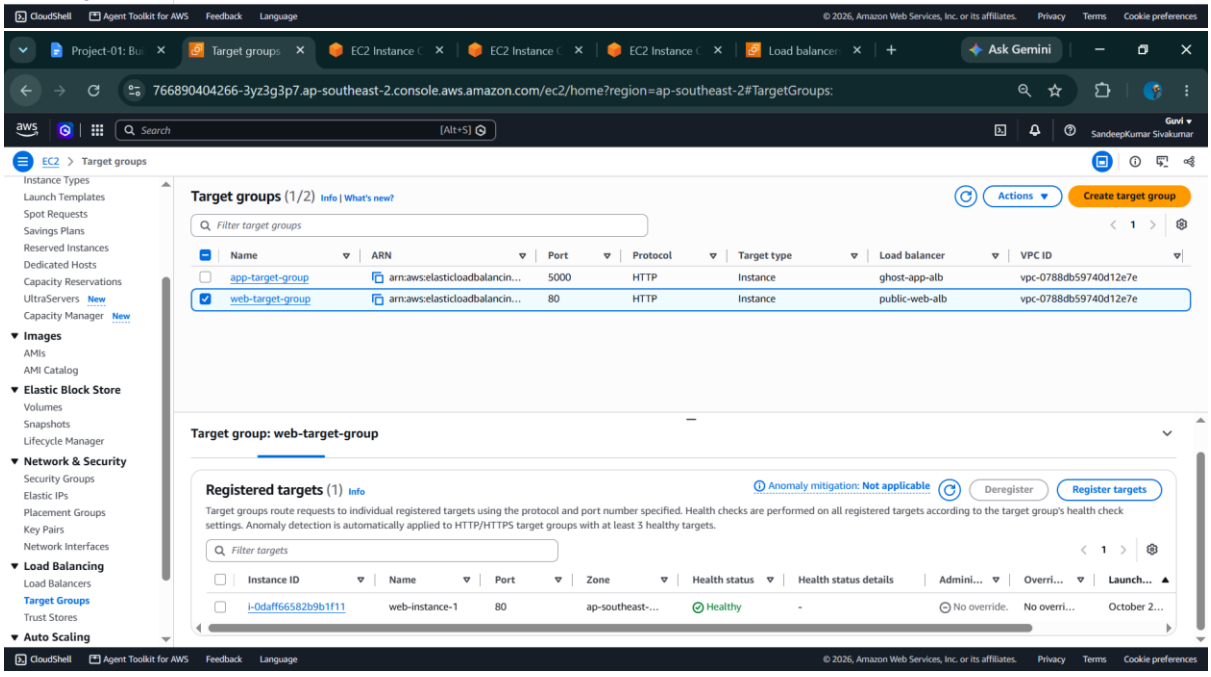


Task 5: Implement Network and Instance Security

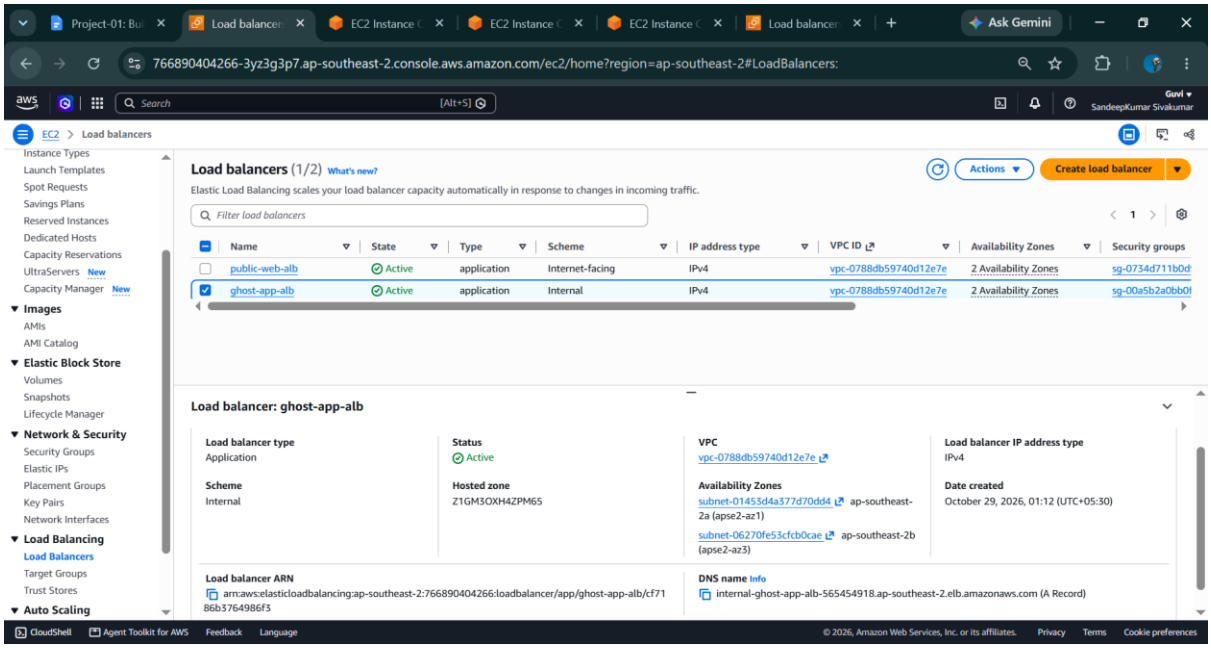


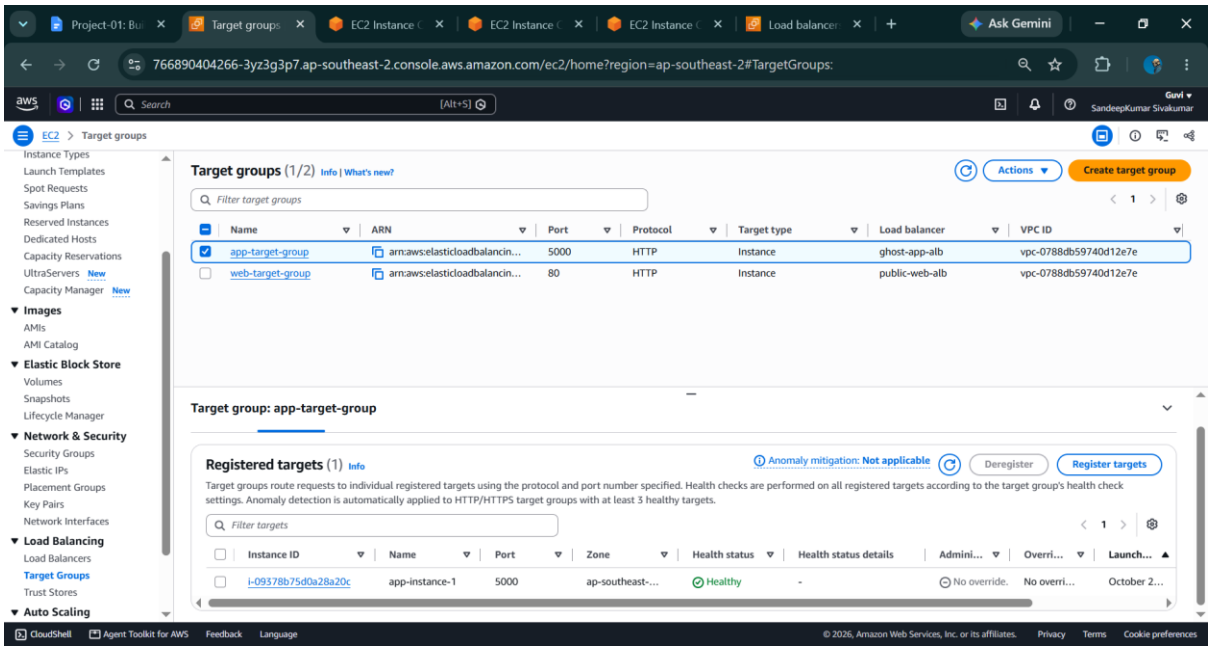
Task 6: Configure the Internet-Facing Application Load Balancer



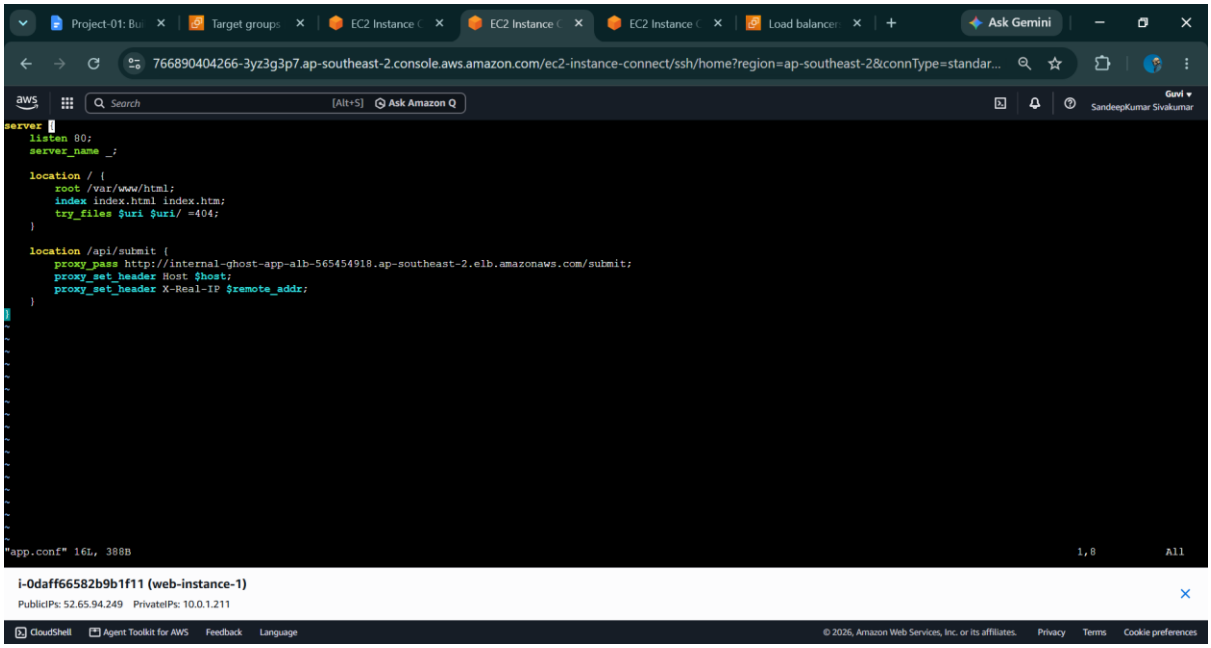


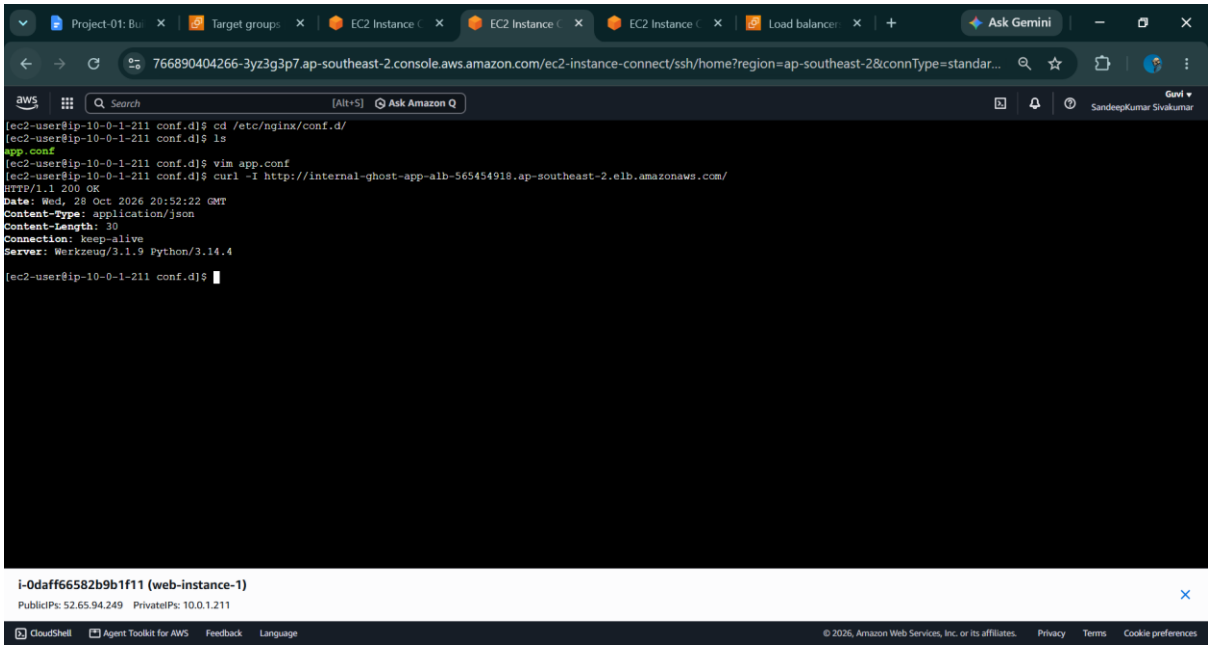
Task 7: Configure Internal Application Load Balancer



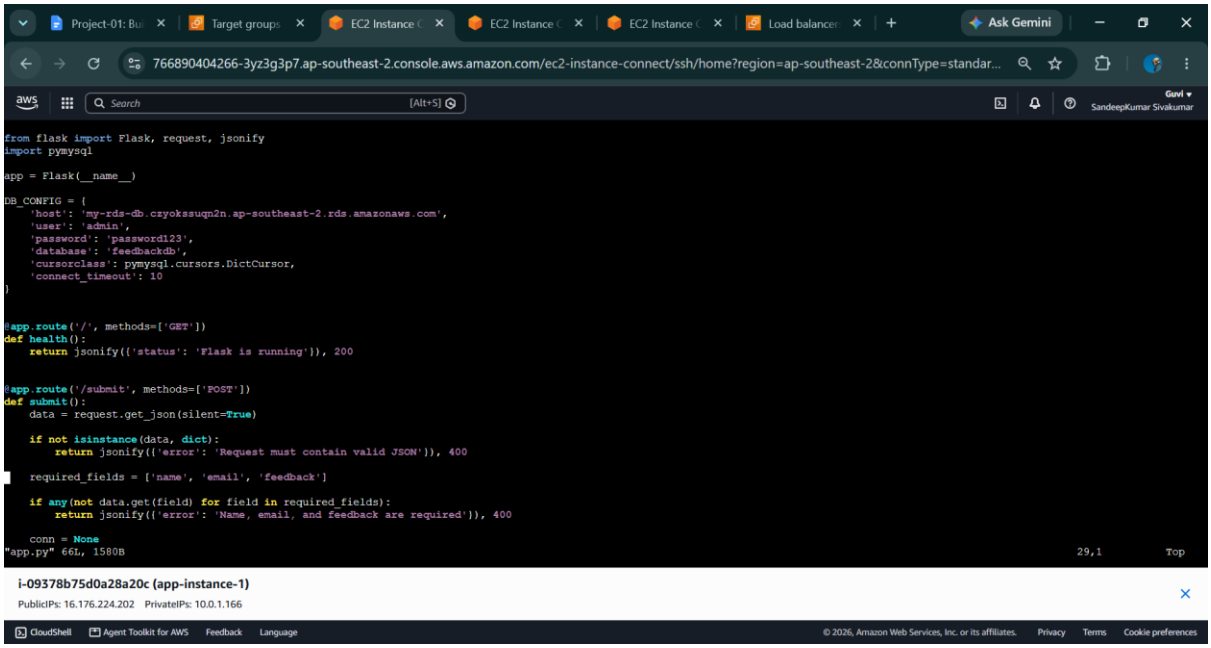


Task 8: Configure Nginx Reverse Proxy for API Routing





Task 9: Integrate the Flask Application with Amazon RDS



```
from flask import Flask, request, jsonify
import pymysql

app = Flask(__name__)

DB_CONFIG = {
    'host': 'my-rds-db.czyokssuqn2n.ap-southeast-2.rds.amazonaws.com',
    'user': 'admin',
    'password': 'password123',
    'database': 'feedbackdb',
    'cursorclass': pymysql.cursors.DictCursor,
    'connect_timeout': 10
}


@app.route('/', methods=['GET'])
def health():
    return jsonify({'status': 'Flask is running'}), 200


@app.route('/submit', methods=['POST'])
def submit():
    data = request.get_json(silent=True)

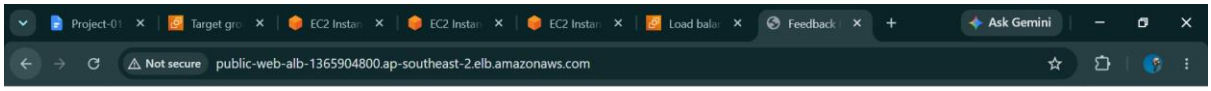
    if not isinstance(data, dict):
        return jsonify({'error': 'Request must contain valid JSON'}), 400

    required_fields = ['name', 'email', 'feedback']

    if any(not data.get(field) for field in required_fields):
        return jsonify({'error': 'Name, email, and feedback are required'}), 400

    conn = None
"app.py" 66L, 1580B                                                    29,1          Top
```

i-09378b75d0a28a20c (app-instance-1)
PublicIPs: 16.176.224.202 PrivateIPs: 10.0.1.166



Feedback Form

guviproj1

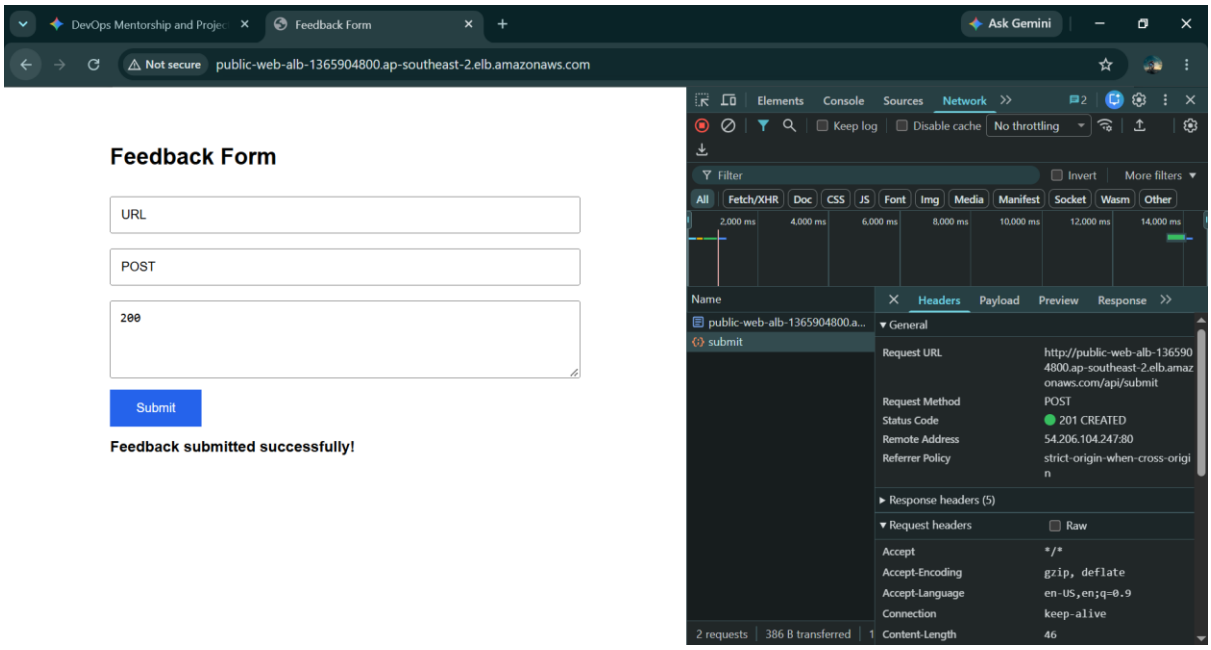
san@guvi.com

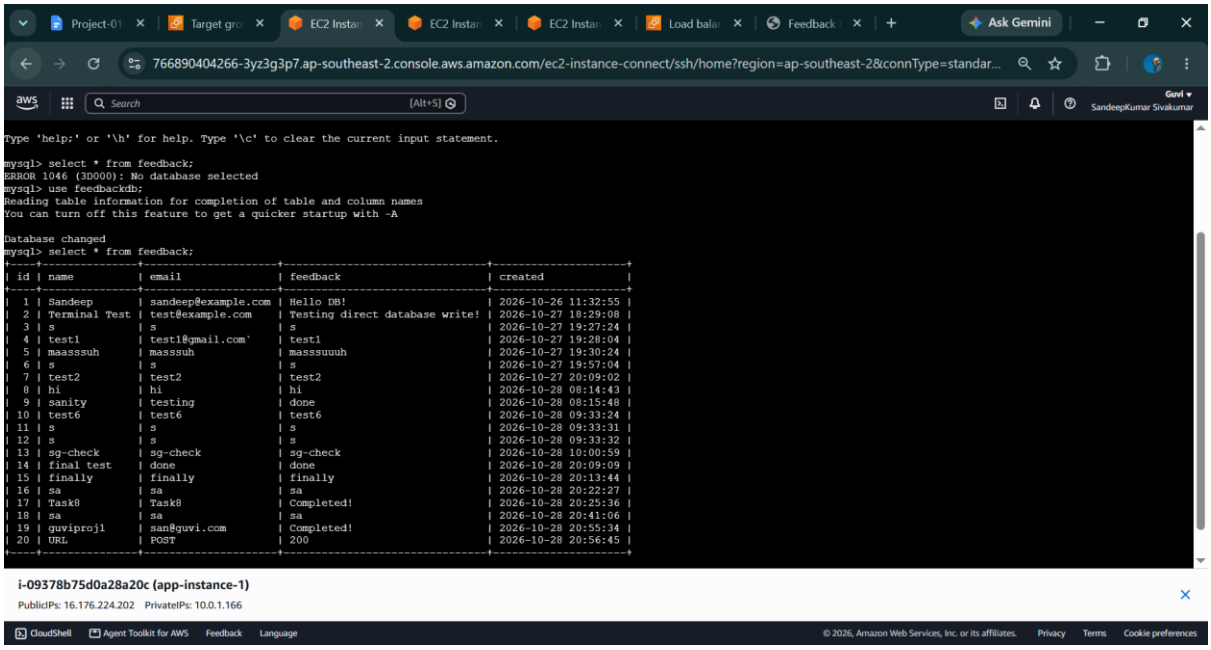
Completed!

Submit

Feedback submitted successfully!

Task 10: Validate the Complete 3-Tier Architecture



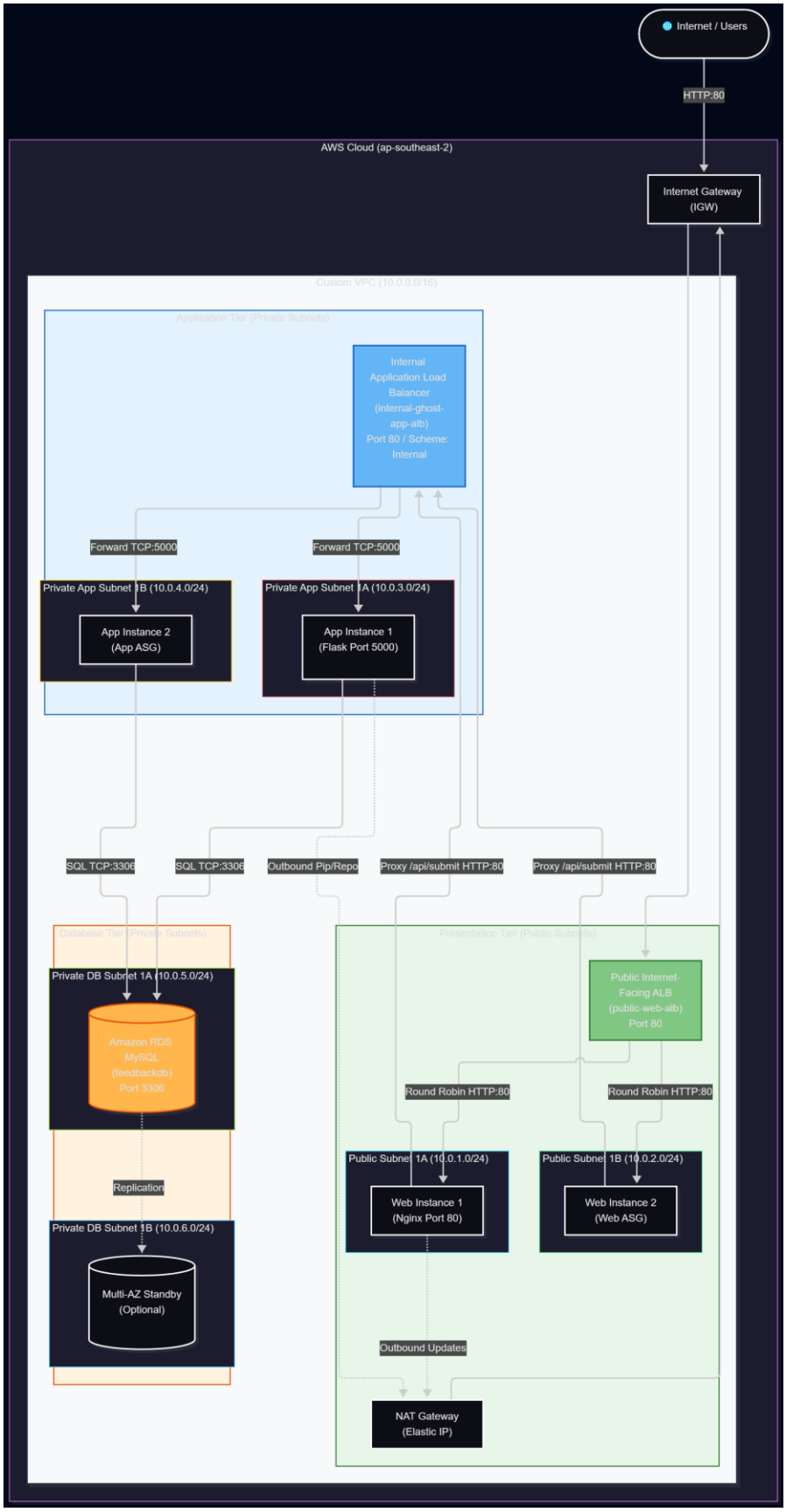


SQL query to view all records from the feedback table:

USE feedbackdb;

SELECT * FROM feedback;

Architecture Diagram:



Done By: **SandeepKumar Sivakumar** Batch: **DO-C-WD-E-B72**